Department of Computer Science
Vrije Universiteit Amsterdam

Resit Exam for *Logic and Sets*
June 3, 2021, 18:45–21:30

This exam has 2 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

By participating in this exam, you underwrite the following `Statement of Integrity`:

I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to make this exam completely on my own, only consult those sources that are allowed explicitly, not share my solutions with other students, and make myself available for any oral clarifications regarding this exam.

1. **Sheffer stroke** (10 points)
   Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

2. **DPLL procedure** (12 points)
   Apply the DPLL procedure to the CNF
   $$(\neg p \vee q \vee \neg r) \ \wedge \ (\neg p \vee \neg q \vee \neg r) \ \wedge \ (p \vee \neg q \vee \neg r) \ \wedge \ r$$
   to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p, then on q, and then on r; and let it always try the truth value true ($\top$) before false ($\bot$).

3. **OBDD** (12 points)
   Represent $z \to (x \oplus y)$ by means of a binary decision tree, with respect to the variable ordering x, y, z. Moreover, reduce this binary decision tree to an ordered binary decision diagram. (Also give the intermediate reduction steps.)

4. **Predicate logic** (11 points)
   Give two models to show that the formulas $C(x) \to D(x)$ and $D(x) \to C(x)$ do not semantically entail each other. (Explain your answer.)

5. **Sets** (6+6 points)
   Consider the following set-theoretic equality
   $$(((A' \cap C) \cup C') \cap B)' \ = \ B' \cup (A \cap C).$$
   (a) Prove it using the rules of the algebra of sets.
   (b) Assuming that $\#U = 14$, $\#A = 5$, $\#B = 7$, $\#C = 6$, $\#(A \cup C) = 9$, and $\#(A \cap B' \cap C) = 1$, compute the number of elements in $B' \cup (A \cap C)$.

6. **Relations** (3+4+4 points)
In the set $V = \{a, b, c, d\}$ consider the relations

$$\begin{aligned} R &= \{\langle a,a\rangle, \langle b,a\rangle, \langle b,b\rangle, \langle b,d\rangle, \langle c,c\rangle, \langle d,c\rangle, \langle d,d\rangle\}, \\ S &= \{\langle a,b\rangle, \langle d,c\rangle\}. \end{aligned}$$

   (a) Write down the matrix representation of the relation R.
   (b) Is the relation R reflexive? Is it transitive? Is it an ordering relation?
   (c) Determine the relation $S \circ R$ by listing all its elements in the curly-bracket notation.

7. **Functions** (5+6 points)
We are given the functions $sqr : \mathbb{R} \to \mathbb{R}$, $mul_3 : \mathbb{R} \to \mathbb{R}$, $sub_2 : \mathbb{R} \to \mathbb{R}$ defined by

$$sqr(x) = x^2, \qquad mul_3(x) = 3x, \qquad sub_2(x) = x - 2.$$

   (a) For each function, determine whether or not the function is injective.
   (b) Express the function $g : \mathbb{R} \to \mathbb{R}$ defined by

$$g(x) = \frac{1}{3}x^2 + \frac{2}{3}$$

   as a composition of sqr, mul_3, sub_2, and their inverses.

8. **Induction and recursion** (3+8 points)
Consider the sequence $(t_n)_{n\in\mathbb{N}}$ of numbers defined recursively by

$$t_0 := \frac{1}{2}, \qquad t_{n+1} := t_n + 3^n + \frac{1}{2}.$$

We claim that the following statement is true for all natural numbers n:

$$t_n = \frac{1}{2}\left(3^n + n\right).$$

   (a) Verify by explicit computation that the claim is true for $n = 0$, $n = 1$ and $n = 2$.
   (b) Prove by mathematical induction that the statement holds true for all natural numbers n.

After completing the exam, show your solutions to the camera before closing Proctorio.

After closing Proctorio, upload your solutions on Canvas, within 15 minutes.